

brain-drain bay card: ASM1153E USB-SATA bridge, switched 12 V / 5 V, 22-pin SATA receptacle

bridge

ASM1153E USB 3 to SATA bridge, 30 MHz crystal, core switcher, AC coupling

File: sheets/bridge.kicad_sch

bay-switch

Switched 12 V and 5 V for the drive: P-FET high-side switches with soft-start, PTC fuses

File: sheets/bay-switch.kicad_sch

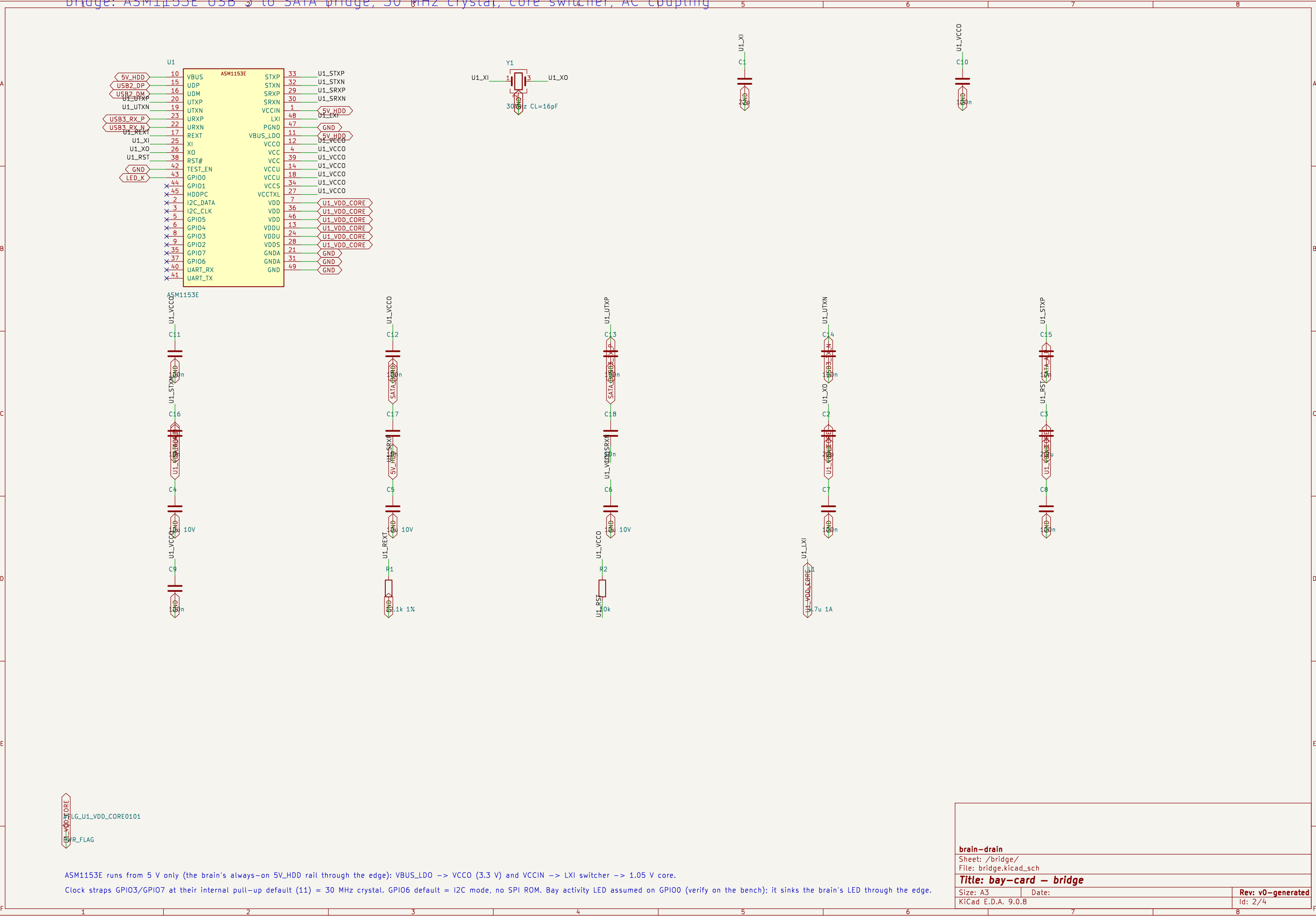
edge

Card edge to the brain (PCIe x1 fingers, bay-card pinout) and the SATA 22-pin receptacle

File: sheets/edge.kicad_sch

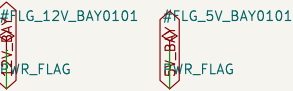
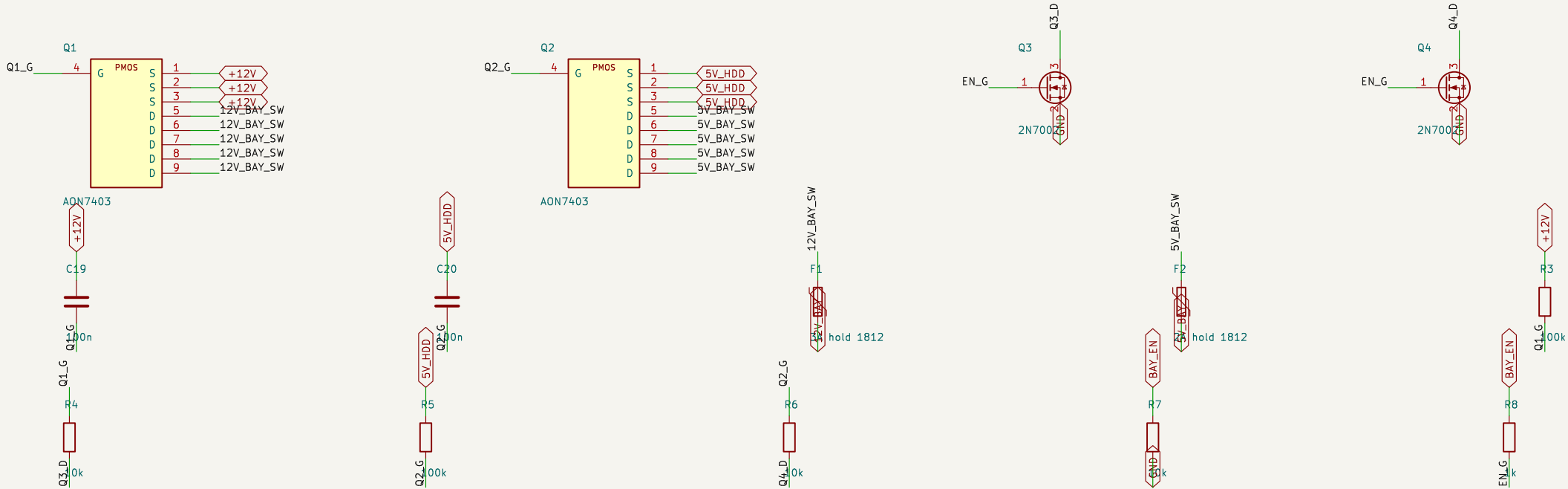
brain-drain		
Sheet: /		
File: bay-card.kicad_sch		
Title: brain-drain bay card: ASM1153E USB-SATA bridge, switched 12 V / 5		
Size: A3	Date:	Rev: v0-generated
KiCad E.D.A. 9.0.8	Id: 1/4	

bridge: ASM1153E USB 3 to SATA bridge, 30 MHz crystal, core switcher, AC coupling



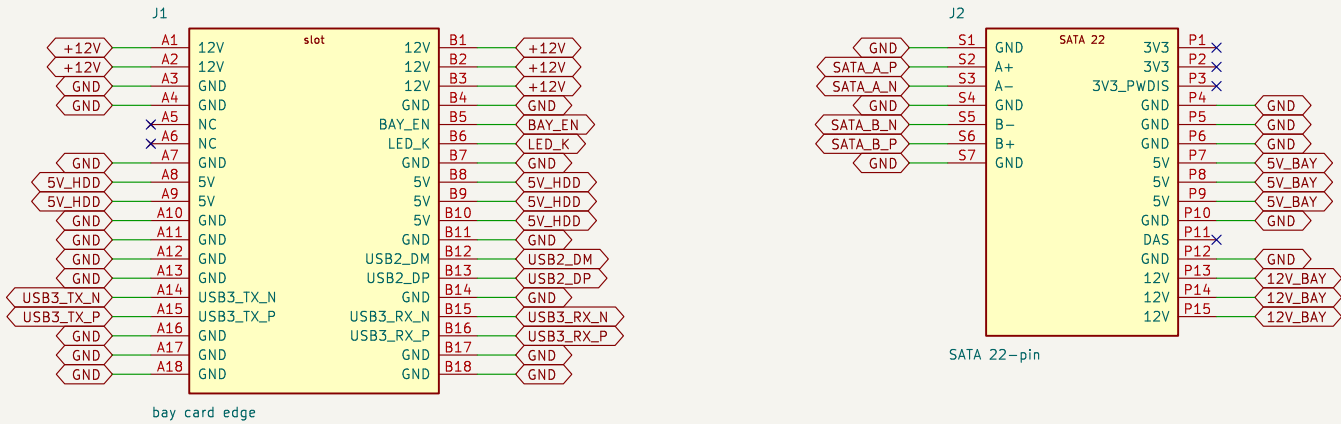
ASM1153E runs from 5 V only (the brain's always-on 5V_HDD rail through the edge): VBUS_LDO -> VCCO (3.3 V) and VCCIN -> LXI switcher -> 1.05 V core.
Clock straps GPIO3/GPIO7 at their internal pull-up default (11) = 30 MHz crystal. GPIO6 default = I2C mode, no SPI ROM. Bay activity LED assumed on GPIO0 (verify on the bench); it sinks the brain's LED through the edge.

bay-switch: Switched 12 V and 5 V for the drive: P-FET high-side switches with soft-start, PTC fuses



P-FET high-side switch; 100k gate pull-up holds it off; the 2N7002 pulls the gate down through 10k with 100n gate-source, ~1 ms ramp so the drive's bulk caps do not trip the fuse. BAY_EN pull-down keeps the bay off through bay

brain-drain		
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Size: A3	Date:	Rev: v0-generated
KiCad E.D.A. 9.0.8		Id: 3/4



Fingers: Connector_PCBEde:BUS_PCIexpress_x1 on the card's bottom edge (chamfer the edge, ENIG is fine for a few insertions). Pinout in symgen.SLOT_PINS.
SATA P1-P3 (3.3 V) are not connected on purpose: P3 is PWDIS on SATA 3.3 drives and a 3.3 V supply there keeps them from spinning up.